

MA388 Sabermetrics: Lesson 21

Simulating Baseball - Part I

Lesson Objectives

1. Construct a transition probability matrix from Retrosheet play-by-play data.
2. Model a baseball half-inning as a Markov chain and simulate 10,000 half-innings.
3. Compute the distribution of runs scored per half-inning and the average runs per half-inning.
4. Discuss the memoryless property of Markov chains and whether it allows for momentum.

Markov Chains

Markov chains model the probabilities of moving between different states. I want you to draw a state diagram (think a network diagram with states as circles and lines connecting the states with probabilities) for a baseball half-inning based on the number of outs where probabilities reflect the state transitions after one plate appearance. Assume that every plate appearance has a .7 probability of resulting in an out and .3 probability of not and there is never more than one out for a plate appearance.

What is the probability of having 0 outs after two plate appearances? What about having two outs?

What is the memoryless property of Markov chains? Does this allow for “momentum”?

So this was all very basic, but hopefully a useful intro. What if we wanted to model baseball this way to determine how many at-bats occur in a given baseball half-inning (or entire game)? How might you approach this using Markov Chains and the data we have? How would you change the states?

```
library(tidyverse)
library(knitr)
library(baseballr)

# Load the data.
retro2024 <-
retrosheet_data(years_to_acquire = '2024') |>
  pluck('2024') |>
  pluck('events')
```

Let’s add states, etc., similarly to how we’ve done it in previous lessons. One important difference is that we don’t care if/where baserunners were left on base when the third out was made. The third out is, however, an important state for us to capture.

```

# Add states, etc.
retro2024 <- retro2024 |>
  mutate(runs_before = away_score_ct + home_score_ct,
         half_inning = paste(game_id, inn_ct, bat_home_id),
         runs_scored =
           (bat_dest_id > 3) + (run1_dest_id > 3) +
           (run2_dest_id > 3) + (run3_dest_id > 3))

half_innings <- retro2024 |>
  summarize(outs_inning = sum(event_outs_ct),
            runs_inning = sum(runs_scored),
            runs_start = first(runs_before),
            max_runs = runs_inning + runs_start, .by = 'half_inning')

retro2024_complete <- retro2024 |>
  inner_join(half_innings, by = "half_inning") |>
  mutate(runs_roi = max_runs - runs_before) |>
  mutate(bases =
    paste(ifelse(base1_run_id > '', 1, 0),
          ifelse(base2_run_id > '', 1, 0),
          ifelse(base3_run_id > '', 1, 0), sep = ""),
         state = paste(bases, outs_ct)) |>
  mutate(is_runner1 =
    as.numeric(run1_dest_id == 1 | bat_dest_id == 1),
         is_runner2 =
    as.numeric(run1_dest_id == 2 | run2_dest_id == 2 |
              bat_dest_id == 2),
         is_runner3 =
    as.numeric(run1_dest_id == 3 | run2_dest_id == 3 |
              run3_dest_id == 3 | bat_dest_id == 3),
         new_outs = outs_ct + event_outs_ct,
         new_bases = paste(is_runner1, is_runner2, is_runner3, sep = ""),
         new_state = paste(new_bases, new_outs)) |>
  filter((state != new_state) | (runs_scored > 0)) |>
  filter(outs_inning == 3, bat_event_fl == TRUE) |>
  mutate(new_state = str_replace(new_state, "[0-1]{3} 3", "3"))

```

Let's look at our resulting data frame:

```
retro2024_complete |>
  select(game_id, bat_id, state, new_state) |>
  head(10) |>
  kable()
```

game_id	bat_id	state	new_state
ANA202404050	duraj001	000 0	000 1
ANA202404050	dever001	000 1	100 1
ANA202404050	stort001	100 1	100 2
ANA202404050	yoshm002	100 2	3
ANA202404050	renda001	000 0	000 1
ANA202404050	schan001	000 1	000 2
ANA202404050	troum001	000 2	3
ANA202404050	oneit001	000 0	000 0
ANA202404050	casat001	000 0	000 0
ANA202404050	valde002	000 0	000 1

We're now ready to compute a matrix of transition probabilities. How many starting states are there? What about ending states? How can we compute these state-to-state probabilities?

```
T_matrix <- retro2024_complete |>
  select(state, new_state) |>
  table()
T_matrix
```

	new_state											
state	000 0	000 1	000 2	001 0	001 1	001 2	010 0	010 1	010 2	011 0	011 1	011 2
000 0	1381	30649	0	210	0	0	1983	0	0	0	0	0
000 1	0	961	22729	0	145	0	0	1390	0	0	0	0
000 2	0	0	725	0	0	87	0	0	1154	0	0	0
001 0	13	80	4	2	175	0	20	1	0	0	0	0
001 1	0	50	340	0	5	615	0	84	10	0	0	0
001 2	0	0	67	0	0	13	0	0	97	0	0	0
010 0	76	8	18	13	788	0	146	1460	0	14	0	0
010 1	0	139	24	0	20	905	0	226	2398	0	14	0
010 2	0	0	203	0	0	37	0	0	344	0	0	2

011 0	22	1	0	2	91	2	44	53	2	3	251	0
011 1	0	26	3	0	8	235	0	97	125	0	6	469
011 2	0	0	55	0	0	13	0	0	73	0	0	0
100 0	293	5	1111	43	31	0	120	672	0	358	0	0
100 1	0	402	7	0	68	27	0	137	749	0	369	0
100 2	0	0	351	0	0	70	0	0	237	0	0	297
101 0	30	2	50	5	3	10	10	37	0	35	22	0
101 1	0	48	1	0	11	8	0	19	92	0	80	46
101 2	0	0	72	0	0	6	0	0	55	0	0	58
110 0	74	1	0	12	3	206	25	2	20	94	227	0
110 1	0	140	2	0	18	3	0	64	11	0	158	235
110 2	0	0	146	0	0	34	0	0	117	0	0	140
111 0	26	1	0	4	0	48	6	2	3	21	45	9
111 1	0	36	1	0	12	1	0	20	5	0	54	86
111 2	0	0	66	0	0	14	0	0	51	0	0	33

new_state													
state	100 0	100 1	100 2	101 0	101 1	101 2	110 0	110 1	110 2	111 0	111 1	111 2	
000 0	10248	0	0	0	0	0	0	0	0	0	0	0	
000 1	0	7558	0	0	0	0	0	0	0	0	0	0	
000 2	0	0	6070	0	0	0	0	0	0	0	0	0	
001 0	63	2	0	52	0	0	0	0	0	0	0	0	
001 1	0	296	55	0	214	0	0	0	0	0	0	0	
001 2	0	0	337	0	0	347	0	0	0	0	0	0	
010 0	167	32	0	298	0	0	410	0	0	0	0	0	
010 1	0	301	47	0	325	0	0	633	0	0	0	0	
010 2	0	0	598	0	0	178	0	0	963	0	0	0	
011 0	35	3	0	47	2	0	5	4	0	68	0	0	
011 1	0	104	5	0	110	43	0	10	16	0	244	0	
011 2	0	0	176	0	0	56	0	0	0	0	0	303	
100 0	1	4640	0	385	0	0	1964	0	0	0	0	0	
100 1	0	1	5855	0	506	0	0	2306	0	0	0	0	
100 2	0	0	10	0	0	631	0	0	2102	0	0	0	
101 0	0	154	2	34	237	0	127	18	0	64	0	0	
101 1	0	0	322	0	68	472	0	244	34	0	151	0	
101 2	0	0	1	0	0	137	0	0	203	0	0	189	
110 0	0	1	17	65	257	0	101	986	0	447	0	0	
110 1	0	0	14	0	146	448	0	171	1726	0	624	0	
110 2	0	0	2	0	0	255	0	0	214	0	0	668	
111 0	0	3	2	15	73	4	23	49	1	108	235	0	
111 1	0	0	4	0	33	144	0	51	128	0	217	493	
111 2	0	0	0	0	0	99	0	0	75	0	0	236	

new_state	
state	3
000 0	0
000 1	0
000 2	18355

```

001 0    0
001 1    11
001 2  1817
010 0    0
010 1    20
010 2  4419
011 0    0
011 1    30
011 2  1315
100 0    0
100 1  1283
100 2  8023
101 0     1
101 1   192
101 2  1544
110 0     1
110 1   483
110 2  3656
111 0     0
111 1   161
111 2  1280

```

```

P_matrix <- prop.table(T_matrix, 1)
dim(P_matrix)

```

```
[1] 24 25
```

Once you get to three outs, there's nowhere left to go – you will stay in three outs with probability 1.

```

P_matrix <- P_matrix |>
  rbind("3" = c(rep(0, 24), 1))
# dim(P_matrix)
P_matrix |> round(3)

```

```

      000 0 000 1 000 2 001 0 001 1 001 2 010 0 010 1 010 2 011 0 011 1 011 2
000 0 0.031 0.689 0.000 0.005 0.000 0.000 0.045 0.000 0.000 0.000 0.000 0.000
000 1 0.000 0.029 0.693 0.000 0.004 0.000 0.000 0.042 0.000 0.000 0.000 0.000
000 2 0.000 0.000 0.027 0.000 0.000 0.003 0.000 0.000 0.044 0.000 0.000 0.000
001 0 0.032 0.194 0.010 0.005 0.425 0.000 0.049 0.002 0.000 0.000 0.000 0.000
001 1 0.000 0.030 0.202 0.000 0.003 0.366 0.000 0.050 0.006 0.000 0.000 0.000
001 2 0.000 0.000 0.025 0.000 0.000 0.005 0.000 0.000 0.036 0.000 0.000 0.000
010 0 0.022 0.002 0.005 0.004 0.230 0.000 0.043 0.426 0.000 0.004 0.000 0.000
010 1 0.000 0.028 0.005 0.000 0.004 0.179 0.000 0.045 0.475 0.000 0.003 0.000

```

010	2	0.000	0.000	0.030	0.000	0.000	0.005	0.000	0.000	0.051	0.000	0.000	0.000											
011	0	0.035	0.002	0.000	0.003	0.143	0.003	0.069	0.083	0.003	0.005	0.395	0.000											
011	1	0.000	0.017	0.002	0.000	0.005	0.153	0.000	0.063	0.082	0.000	0.004	0.306											
011	2	0.000	0.000	0.028	0.000	0.000	0.007	0.000	0.000	0.037	0.000	0.000	0.000											
100	0	0.030	0.001	0.115	0.004	0.003	0.000	0.012	0.070	0.000	0.037	0.000	0.000											
100	1	0.000	0.034	0.001	0.000	0.006	0.002	0.000	0.012	0.064	0.000	0.032	0.000											
100	2	0.000	0.000	0.030	0.000	0.000	0.006	0.000	0.000	0.020	0.000	0.000	0.025											
101	0	0.036	0.002	0.059	0.006	0.004	0.012	0.012	0.044	0.000	0.042	0.026	0.000											
101	1	0.000	0.027	0.001	0.000	0.006	0.004	0.000	0.011	0.051	0.000	0.045	0.026											
101	2	0.000	0.000	0.032	0.000	0.000	0.003	0.000	0.000	0.024	0.000	0.000	0.026											
110	0	0.029	0.000	0.000	0.005	0.001	0.081	0.010	0.001	0.008	0.037	0.089	0.000											
110	1	0.000	0.033	0.000	0.000	0.004	0.001	0.000	0.015	0.003	0.000	0.037	0.055											
110	2	0.000	0.000	0.028	0.000	0.000	0.006	0.000	0.000	0.022	0.000	0.000	0.027											
111	0	0.038	0.001	0.000	0.006	0.000	0.071	0.009	0.003	0.004	0.031	0.066	0.013											
111	1	0.000	0.025	0.001	0.000	0.008	0.001	0.000	0.014	0.003	0.000	0.037	0.059											
111	2	0.000	0.000	0.036	0.000	0.000	0.008	0.000	0.000	0.028	0.000	0.000	0.018											
3		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
	100	0	100	1	100	2	101	0	101	1	101	2	110	0	110	1	110	2	111	0	111	1	111	2
000	0	0.230	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
000	1	0.000	0.231	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
000	2	0.000	0.000	0.230	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
001	0	0.153	0.005	0.000	0.126	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
001	1	0.000	0.176	0.033	0.000	0.127	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
001	2	0.000	0.000	0.126	0.000	0.000	0.130	0.000	0.000	0.000	0.000	0.000	0.000											
010	0	0.049	0.009	0.000	0.087	0.000	0.000	0.120	0.000	0.000	0.000	0.000	0.000											
010	1	0.000	0.060	0.009	0.000	0.064	0.000	0.000	0.125	0.000	0.000	0.000	0.000											
010	2	0.000	0.000	0.089	0.000	0.000	0.026	0.000	0.000	0.143	0.000	0.000	0.000											
011	0	0.055	0.005	0.000	0.074	0.003	0.000	0.008	0.006	0.000	0.107	0.000	0.000											
011	1	0.000	0.068	0.003	0.000	0.072	0.028	0.000	0.007	0.010	0.000	0.159	0.000											
011	2	0.000	0.000	0.088	0.000	0.000	0.028	0.000	0.000	0.000	0.000	0.000	0.152											
100	0	0.000	0.482	0.000	0.040	0.000	0.000	0.204	0.000	0.000	0.000	0.000	0.000											
100	1	0.000	0.000	0.500	0.000	0.043	0.000	0.000	0.197	0.000	0.000	0.000	0.000											
100	2	0.000	0.000	0.001	0.000	0.000	0.054	0.000	0.000	0.179	0.000	0.000	0.000											
101	0	0.000	0.183	0.002	0.040	0.282	0.000	0.151	0.021	0.000	0.076	0.000	0.000											
101	1	0.000	0.000	0.180	0.000	0.038	0.264	0.000	0.136	0.019	0.000	0.084	0.000											
101	2	0.000	0.000	0.000	0.000	0.000	0.060	0.000	0.000	0.090	0.000	0.000	0.083											
110	0	0.000	0.000	0.007	0.026	0.101	0.000	0.040	0.388	0.000	0.176	0.000	0.000											
110	1	0.000	0.000	0.003	0.000	0.034	0.106	0.000	0.040	0.407	0.000	0.147	0.000											
110	2	0.000	0.000	0.000	0.000	0.000	0.049	0.000	0.000	0.041	0.000	0.000	0.128											
111	0	0.000	0.004	0.003	0.022	0.108	0.006	0.034	0.072	0.001	0.159	0.347	0.000											
111	1	0.000	0.000	0.003	0.000	0.023	0.100	0.000	0.035	0.089	0.000	0.150	0.341											
111	2	0.000	0.000	0.000	0.000	0.000	0.053	0.000	0.000	0.040	0.000	0.000	0.127											
3		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000											
	3																							
000	0	0.000																						
000	1	0.000																						

000	2	0.696
001	0	0.000
001	1	0.007
001	2	0.678
010	0	0.000
010	1	0.004
010	2	0.655
011	0	0.000
011	1	0.020
011	2	0.660
100	0	0.000
100	1	0.110
100	2	0.684
101	0	0.001
101	1	0.107
101	2	0.682
110	0	0.000
110	1	0.114
110	2	0.699
111	0	0.000
111	1	0.111
111	2	0.690
3		1.000

How can we check the validity of this transition matrix? Pick a state – what should the sum of all transition probabilities from that state to any other be?

```
P_matrix |>
  apply(MARGIN = 1, FUN = sum)
```

```
000 0 000 1 000 2 001 0 001 1 001 2 010 0 010 1 010 2 011 0 011 1 011 2 100 0
      1      1      1      1      1      1      1      1      1      1      1      1      1
100 1 100 2 101 0 101 1 101 2 110 0 110 1 110 2 111 0 111 1 111 2      3
      1      1      1      1      1      1      1      1      1      1      1      1      1
```

Let's examine the nobody-on, nobody-out state. What are the most likely resulting states?

```
P_matrix |>
  as_tibble(rownames = "state") |>
  filter(state == "000 0") |>
  pivot_longer(
    cols = -state,
    names_to = "new_state",
    values_to = "Prob"
  ) |>
  filter(Prob > 0) |>
  kable()
```

state	new_state	Prob
000 0	000 0	0.0310539
000 0	000 1	0.6891907
000 0	001 0	0.0047222
000 0	010 0	0.0445909
000 0	100 0	0.2304423

What about with a runner on 2nd base and two outs?

```
P_matrix |>
  as_tibble(rownames = "state") |>
```

```

filter(state == "010 2") |>
pivot_longer(
  cols = -state,
  names_to = "new_state",
  values_to = "Prob"
) |>
filter(Prob > 0) |>
kable()

```

state	new_state	Prob
010 2	000 2	0.0301008
010 2	001 2	0.0054864
010 2	010 2	0.0510083
010 2	011 2	0.0002966
010 2	100 2	0.0886714
010 2	101 2	0.0263938
010 2	110 2	0.1427936
010 2	3	0.6552491

So far, so good... Now we need to actually simulate the Markov Chain. This involves being in a state and transitioning into possible other states in a probabilistic manner...a large number of times.

The key equation for keeping score is:

$$runs = (N_{runners}^{(before)} + Outs^{(before)} + 1) - (N_{runners}^{(after)} + Outs^{(after)})$$

```

# This function essentially "adds" the state: "011 1" becomes 3.
num_havent_scored <- function(s) {
  s |>
  str_split("") |>
  pluck(1) |>
  as.numeric() |>
  sum(na.rm = TRUE)
}

# "Add" the state (runners + outs) for every state.
runners_out <- T_matrix |>
row.names() |>
set_names() |>
map_int(num_havent_scored)

# This operation computes the runs equation above for every state-to-state

```

```
# transition, including the impossible transitions that we can safely ignore.
R_runs <- outer(
  runners_out + 1,
  runners_out,
  FUN = "-"
) |>
  cbind("3" = rep(0, 24))
```

Before we go any further, what are the key tasks for simulating a half-inning of baseball?

Here we go - let's simulate!

```
# This function returns the number of runs scored in a simulated half-inning.
simulate_half_inning <- function(P, R, start = 1) {
  s <- start
  path <- NULL
  runs <- 0
  while (s < 25) {
    s_new <- sample(1:25, size = 1, prob = P[s, ])
    path <- c(path, s_new)
    runs <- runs + R[s, s_new]
    s <- s_new
  }
  runs
}
```

Let's simulate 10,000 half-innings.

```
set.seed(388)
simulated_runs <- 1:10000 |>
  map_int(~simulate_half_inning(P_matrix, R_runs))
P_matrix[1,]
```

000 0	000 1	000 2	001 0	001 1	001 2
0.031053945	0.689190709	0.000000000	0.004722178	0.000000000	0.000000000
010 0	010 1	010 2	011 0	011 1	011 2
0.044590857	0.000000000	0.000000000	0.000000000	0.000000000	0.000000000
100 0	100 1	100 2	101 0	101 1	101 2

```

0.230442311 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
      110 0      110 1      110 2      111 0      111 1      111 2
0.000000000 0.000000000 0.000000000 0.000000000 0.000000000 0.000000000
      3
0.000000000

```

What was the observed distribution of half-inning run totals?

```
table(simulated_runs)
```

```

simulated_runs
 0    1    2    3    4    5    6    7    8
7435 1411  653  279  120   59   21   17   5

```

How often do teams score at least five runs in an inning?

```
sum(simulated_runs >= 5) / 10000
```

```
[1] 0.0102
```

What is the average number of runs scored per half-inning?

```
mean(simulated_runs)
```

```
[1] 0.4614
```

What have we left out? Name some non-batting plays that have the potential to change the state and potentially score runs.